Lab 2 — Creating Lines and Coordinates

AutoCAD Masterclass (C391) · Topic 02: Technical Drawing with AutoCAD

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Objective

LO2 — produce 2D technical drawings using AutoCAD

Goal

Draw the outline of practice drawing EX-1 using the LINE and PLINE commands with absolute, relative and polar coordinate entry, working from its dimensioned drawing sheet.

You'll produce

The completed EX-1 profile drawn to size, matching the supplied solution DWG.

Commands / tools: LINE, PLINE, coordinate entry (@dx,dy · @dist<angle), ORTHO, dynamic input

Files in this lab

- ex1.dwg
- ex1-drawing-sheet.pdf
- ex1.png

Step-by-step

1. Open the EX-1 drawing sheet (ex1-drawing-sheet.pdf) and study the dimensions
2. Start a new acadiso drawing and draw the first edge with absolute coordinates

   ```
   L → 0,0 → 100,0
   ```

3. Continue the profile with relative coordinates for each dimensioned edge

   ```
   @0,50  ·  @-30,0  ·  @20<135
   ```

4. Use ortho (F8) and polar tracking (F10) for the horizontal and vertical segments

   ```
   F8 · F10
   ```

5. Close the profile back to the start point and check it forms one closed loop

   ```
   C (Close)
   ```

6. Compare your result against the solution file ex1.dwg, then save

   ```
   OPEN ex1.dwg → SAVEAS ex1-mine.dwg
   ```

Check your work

Your outline closes exactly (LIST shows the final vertex at 0,0) and overlays the solution ex1.dwg with no visible deviation.

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